TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Determination of Porosity of Fabric

*Cover factors, fractional cover and the open window between the threads — porosity calculated from construction, then checked by the air that actually passes through.*

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 5 · LEVEL 3 TERM 2

## 1. Aim and theory

**Aim:** to determine the porosity of the reference plain-weave fabric from its construction — Peirce cover factors, fractional cover and open-area percentage — and to check the result functionally by an air-permeability measurement to ISO 9237.

### 1.1 Cover factor

A woven fabric is threads and gaps. **Cover** asks how completely the threads hide the plane of the fabric; **porosity**, in the geometric sense used here, is what remains open. Peirce's cover factor combines thread density with yarn fineness:

$$\text{Cover factor } K = \text{threads per inch} / \sqrt{(\text{count, Ne})}$$

The square root appears because yarn diameter is proportional to $1/\sqrt{Ne}$ for cotton-count yarns of the same fibre: doubling the threads per inch doubles cover, but doubling yarn diameter (count four times coarser) also doubles cover, and K captures both in one number. Warp and weft have separate cover factors, $K_w$ and $K_f$, from their own densities and counts.

### 1.2 Fractional cover and open area

For cotton yarns, a cover factor of 28 corresponds to threads just touching — fractional cover 1.0 in that direction. Hence:

$$\text{Fractional cover (one direction)} = K / 28$$

Warp threads cover their fraction of the area and weft threads cover theirs, but the overlap — where a weft crosses a warp — must not be counted twice:

$$\text{Combined fractional cover} = C_w + C_f - C_w \times C_f$$

$$\text{Porosity (open area, \%)} = (1 - \text{combined fractional cover}) \times 100$$

This geometric porosity is the fraction of the fabric area that is open window between threads. It is an idealisation: real yarns are hairy, flattened in the weave and not perfectly spaced, so the true open area is somewhat less than the calculation — which is exactly why the functional check below matters.

### 1.3 The functional check: air permeability (ISO 9237)

ISO 9237 measures the rate of air flow passing perpendicularly through a known fabric area under a stated pressure difference (100 Pa for apparel fabrics is customary, test area commonly 20 cm²). The result is quoted as air permeability in cm³/cm²/s (equivalently mm/s) or L/min for the test area. Air permeability is the porosity the user experiences: wind resistance, breathability and drying all follow it. It depends on the open area, but also on pore shape, fabric thickness (Experiment 3 — the air must travel 0.434 mm) and surface hairiness, so it complements rather than duplicates the geometric calculation.

$$\text{Air permeability (cm}^3\text{/cm}^2\text{/s)} = \text{flow (cm}^3\text{/s)} / \text{test area (cm}^2\text{)}$$

All measurements are made on fabric conditioned at 65% RH and 20 °C.

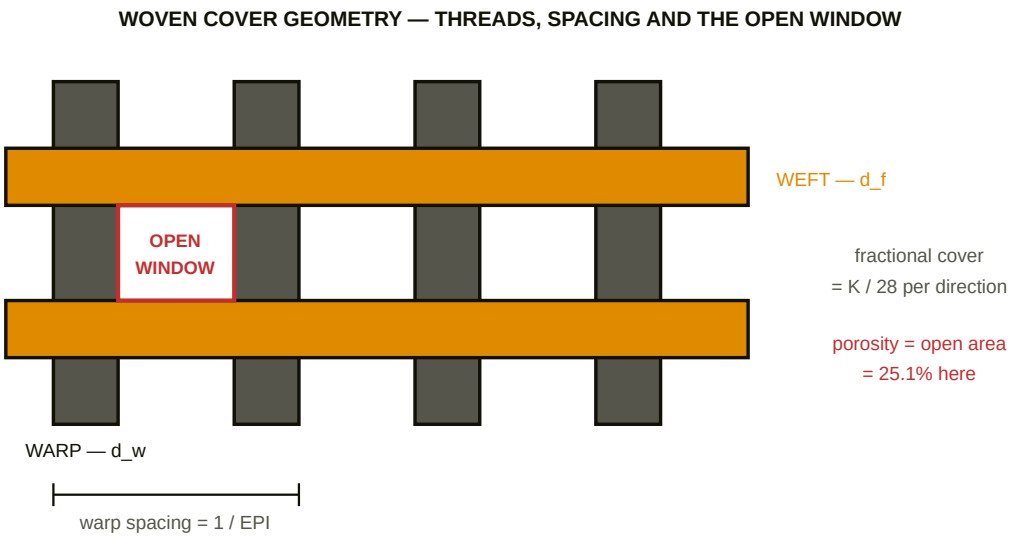



Figure 1 — Plain-weave cover geometry: warp threads (grey) and weft threads (orange) of diameters $d_w$ and $d_f$ at spacings set by EPI and PPI leave a regular grid of open windows; the windows' share of the total area is the geometric porosity.

## 2. Apparatus and procedure

- **Apparatus:** construction data from Experiments 1 and 2 (EPI, PPI, yarn counts from the fabric analysis); air-permeability tester to ISO 9237 with 20 cm² test head and 100 Pa pressure difference; conditioned specimens of the reference fabric.
- **Geometric route:** take warp density 70 EPI with warp count Ne 20 and weft density 59 PPI with weft count Ne 24. Compute each cover factor K = threads per inch / √Ne, convert to fractional cover

by dividing by 28, combine with the overlap correction, and subtract from 1 for porosity.

- **Air-permeability route:** clamp the conditioned specimen flat over the 20 cm² test area without stretching or leakage at the edges, set the pressure difference to 100 Pa, allow the flow to stabilise and read the flow rate. Repeat at ten spread positions and average.
- **Conversion:** convert the mean flow in L/min to cm³/s (× 1000 / 60) and divide by the 20 cm² test area to obtain air permeability in cm³/cm²/s.
- **Comparison:** set the geometric open-area percentage beside the measured air permeability and explain their relationship — same fabric, two views of its openness.

## 3. Worked calculation

**Cover factors.**

$$K_w = 70 / \sqrt{20} = 70 / 4.472 = 15.65$$

$$K_f = 59 / \sqrt{24} = 59 / 4.899 = 12.04$$

**Fractional covers.**

$$C_w = 15.65 / 28 = 0.559 \cdot C_f = 12.04 / 28 = 0.430$$

**Combined cover and porosity.**

$$\text{Combined cover} = 0.559 + 0.430 - (0.559 \times 0.430) = 0.989 - 0.240 = 0.749$$

$$\text{Porosity} = (1 - 0.749) \times 100 = 25.1\% \text{ open area}$$

**Air-permeability check (ISO 9237).** Mean flow at 100 Pa over 20 cm² = 3.36 L/min.

$$\text{Flow} = 3.36 \times 1000 / 60 = 56.0 \text{ cm}^3/\text{s} \cdot \text{Air permeability} = 56.0 / 20 = 2.8 \text{ cm}^3/\text{cm}^2/\text{s}$$

| QUANTITY | VALUE |
|---|---|
| Warp cover factor $K_w$ (70 EPI, Ne 20) | 15.65 |
| Weft cover factor $K_f$ (59 PPI, Ne 24) | 12.04 |
| Combined fractional cover | 0.749 |
| Porosity (open area) | 25.1% |
| Air permeability at 100 Pa (ISO 9237) | 2.8 cm³/cm²/s |

**DO THE TWO ROUTES AGREE?**

> An open area of 25.1% with a measured air permeability of only 2.8 cm³/cm²/s is fully consistent: geometric porosity counts area, but air must squeeze through windows narrowed by yarn hairiness and flattened, crimped yarns, along a 0.434 mm path (Experiment 3). A quarter of the area is window, yet the fabric is still a fairly wind-resistant apparel cloth — cover factor predicts the structure, air permeability grades the performance.

## 4. Precautions

- Use the measured thread densities and the actual yarn counts, not nominal ticket values — cover factor is only as honest as its inputs (here 70 × 59 and Ne 20 / Ne 24).
- Do not add fractional covers without the overlap correction; the crossed area belongs to both thread systems and double-counting overstates cover.
- In the air test, clamp without stretching: tension opens the windows and inflates permeability, and a leaking clamp edge inflates it further.
- State the pressure difference (100 Pa) and test area (20 cm²) with every air-permeability result; values at other pressures are not interchangeable.
- Average at least ten positions: local density variation (Experiment 1) makes single-spot air readings scatter widely.
- Remember the geometric result is an idealisation — report it as calculated open area, and let the ISO 9237 value carry the functional claim.

## Practice (viva preparation)

1. Find the warp cover factor for 70 EPI in Ne 20 yarn, and its fractional cover.
2. Find the weft cover factor for 59 PPI in Ne 24 yarn, and its fractional cover.
3. Combine the two fractional covers of Questions 1–2 and find the porosity. Why is the product term subtracted?
4. An air tester reads 3.36 L/min over 20 cm² at 100 Pa. Find the air permeability in cm³/cm²/s.
5. Why is the √Ne in the cover-factor formula — what yarn property does it represent?
6. A fabric shows 25% open area yet low air permeability. Give two structural reasons.
7. How would porosity change if the same yarns were woven at 60 EPI × 50 PPI instead of 70 × 59?

### Answer key

1. $K_w$ = 70 / √20 = 70 / 4.472 = 15.65. Fractional warp cover = 15.65 / 28 = 0.559.
2. $K_f$ = 59 / √24 = 59 / 4.899 = 12.04. Fractional weft cover = 12.04 / 28 = 0.430.

3. Combined = 0.559 + 0.430 − 0.559 × 0.430 = 0.749. Porosity = 1 − 0.749 = 0.251 = 25.1%. The product term is the overlap: the area where weft crosses warp is covered by both systems, so adding the fractions alone counts it twice; subtracting the product counts every covered square once.
4. 3.36 L/min = 3,360 cm³/min = 56.0 cm³/s. Dividing by the 20 cm² test area: 56.0 / 20 = 2.8 cm³/cm²/s at 100 Pa (ISO 9237).
5. It represents yarn diameter. For yarns of the same fibre and packing, diameter varies as 1/√count (Ne): a yarn four numbers finer in Ne has half the diameter. Cover depends on how much width each thread occupies, so density must be weighted by diameter — which is what dividing by √Ne does (equivalently, K is proportional to threads per inch × diameter).
6. First, yarn hairiness and flattening: protruding fibres and ribbon-like flattened yarns narrow the windows well below their geometric size. Second, path length and pore shape: air must travel through the full fabric thickness (0.434 mm here) through tortuous, slit-like pores, losing energy to friction — area alone overstates flow. Thickness and surface finish therefore separate geometric porosity from measured air permeability.
7. Both cover factors fall: $K_w = 60/\sqrt{20} = 13.4 \rightarrow C_w = 0.479$; $K_f = 50/\sqrt{24} = 10.2 \rightarrow C_f = 0.364$. Combined = 0.479 + 0.364 − 0.174 = 0.669, so porosity rises to about 33.1% — fewer threads of the same size leave bigger windows, and air permeability would rise more than proportionally.

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